Rotary Wing Aerodynamics (AE 667)



Vehicle Performance

Performance Parameters

- Endurance
- Range
- Speed
- Service Altitude
- Payload

Engine Information

We assume that the following information is available from the engine manufacturer

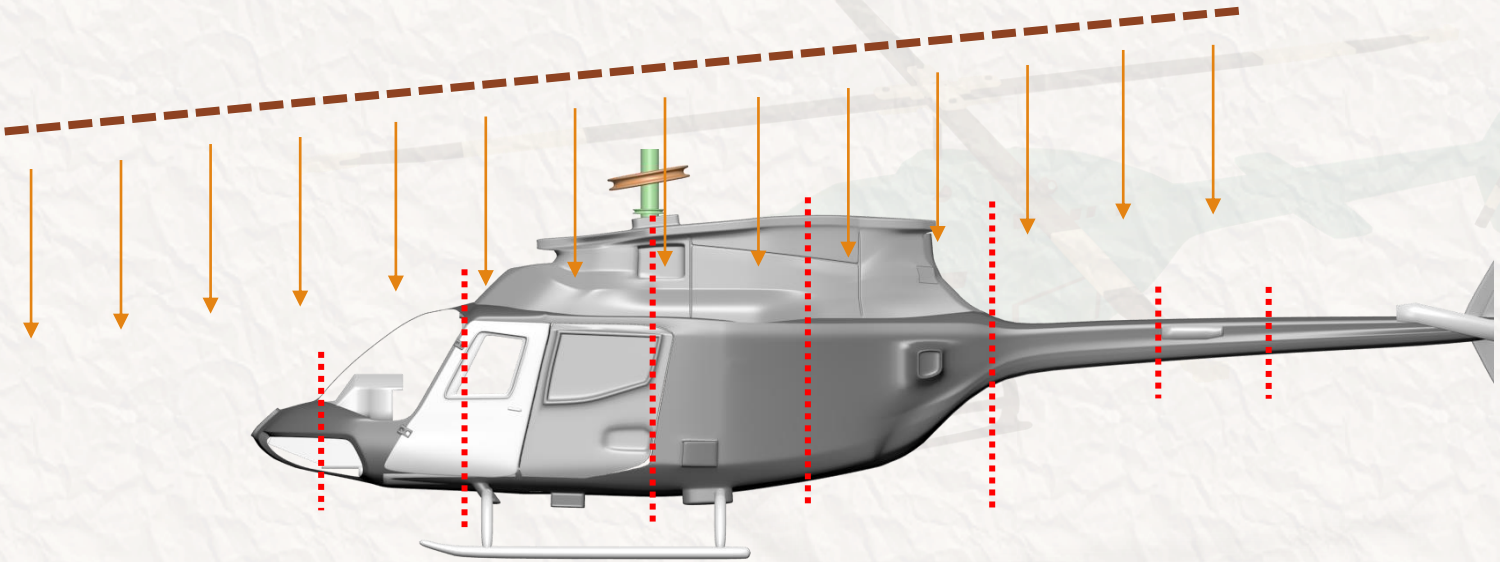
- Shaft power output & Fuel Consumption
- w.r.t: Altitude, Forward speed, Temperature
- at different throttle settings

- Installed power losses: Losses associated with installation of an engine on a vehicle
  (inlet duct skin friction, air filter, exhaust duct skin friction, infrared suppressors, compressor bleed, accessories, transmission losses)



http://www.flight.mace.manchester.ac.uk/air-particle-separator-design/about/about.html

Fuselage Download

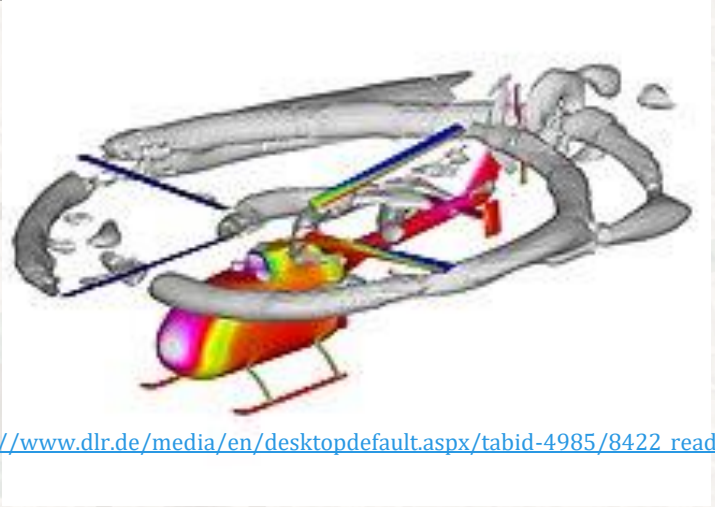


Download: Downward force on fuselage due to rotor downwash

Found experimentally, or by modelling fuselage as a series of segments

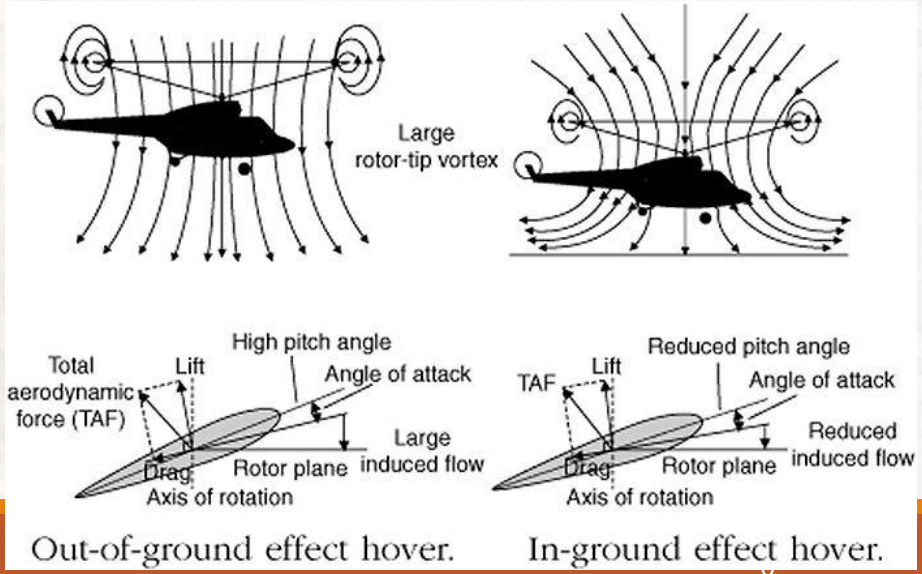
Need to correct for:

- Main rotor – Tail rotor aerodynamic interference [causing higher power requirement at tail rotor]



http://www.dlr.de/media/en/desktopdefault.aspx/tabid-4985/8422_read-19451

- Ground effect [ground restricting rotor inflow, benefiting performance]



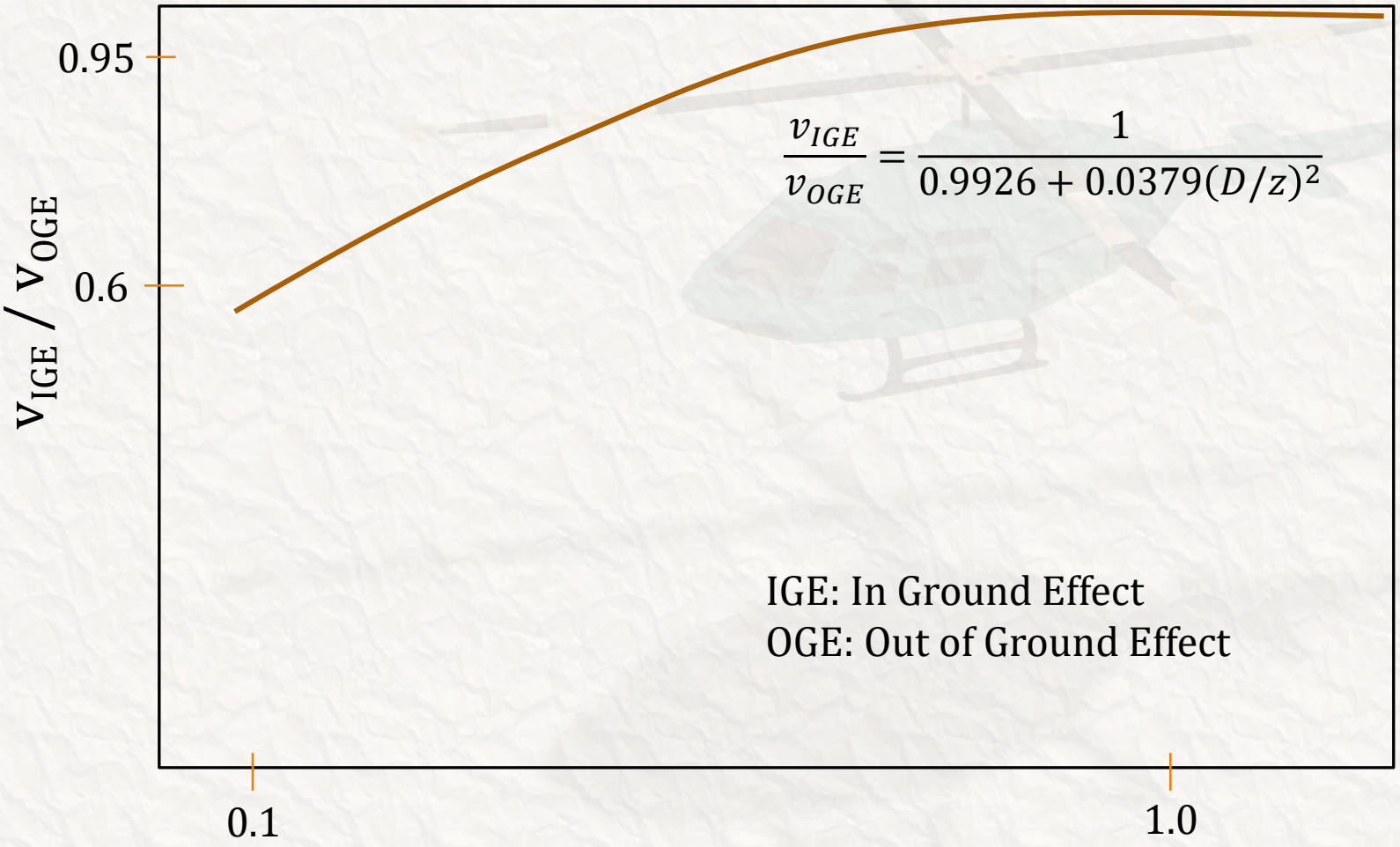
https://www.helis.com/howflies/igeoge.php

Need to correct for:

- Parasitic drag due to fuselage, hub, antennae, pitot probes, hinges, steps, landing gear etc.
- Fuselage drag generally expressed as equivalent flat plate area.



Ground effect in hover



Vertical Climb Performance

- Power requirement includes:
  - Main rotor power
    - Induced power (accounting for thrust = vehicle weight + downwash on fuselage) [P_i]
    - Profile power [P_o]
    - Additional power due to climb [P_c] = climb rate × gross weight
  - Tail rotor power
  - Engine power transmission and installation Losses

- Absolute Ceiling:
  Power required = power available @ Zero climb rate

- Service ceiling:
  Power available = Power required to climb @ 100 ft/min

Power with Altitude

- Available power dependent on density:

$$\frac{P}{P_{sea-level}} = \frac{\rho}{\rho_{sea-level}}$$

- Rotor power's dependency on density

$$P = \kappa T \sqrt{\frac{T}{2\rho A}} + \sigma C_d \, \rho (\Omega R)^3 \pi R^2 / 8$$

(Induced power) (profile power)

Note that induced power increases with decrease in density and profile power decreases with density

- Density dependent on altitude through expression:

$$\frac{\rho}{\rho_{sea-level}} = \left[1 - \frac{0.00198h}{288.16}\right]^{4.2553}$$

Find vertical climb rate at an altitude

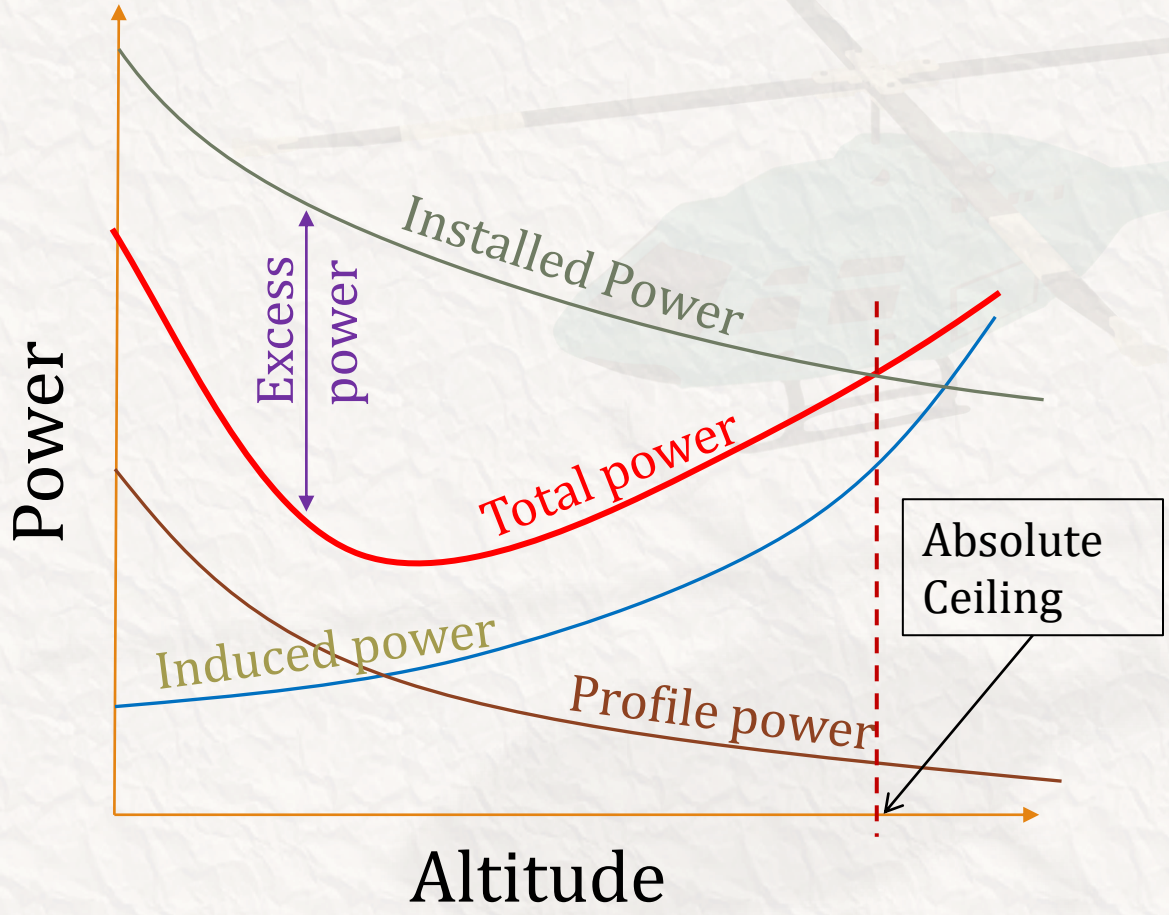
- Find air density ρ at the altitude
- Compute available engine power [installed power]
- Compute required power for hover:

  [Main rotor power (accounting for fuselage download) + Tail rotor power]
- Compute Excess Power

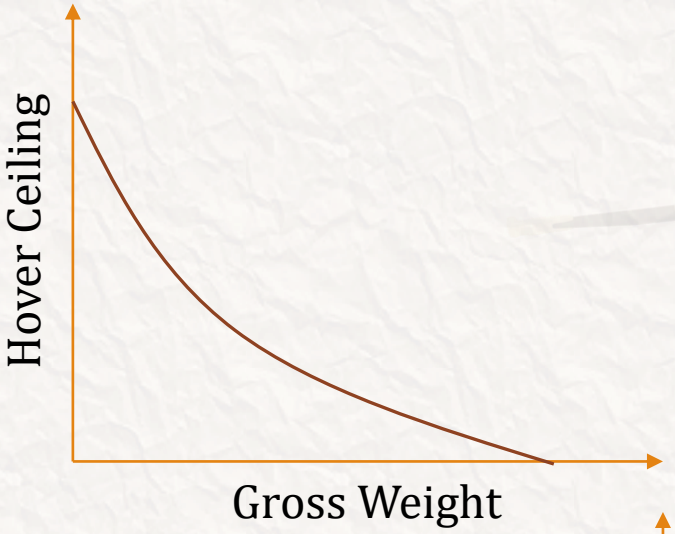
  = Installed power – Required power for hover
- Max climb rate = Excess power/gross weight

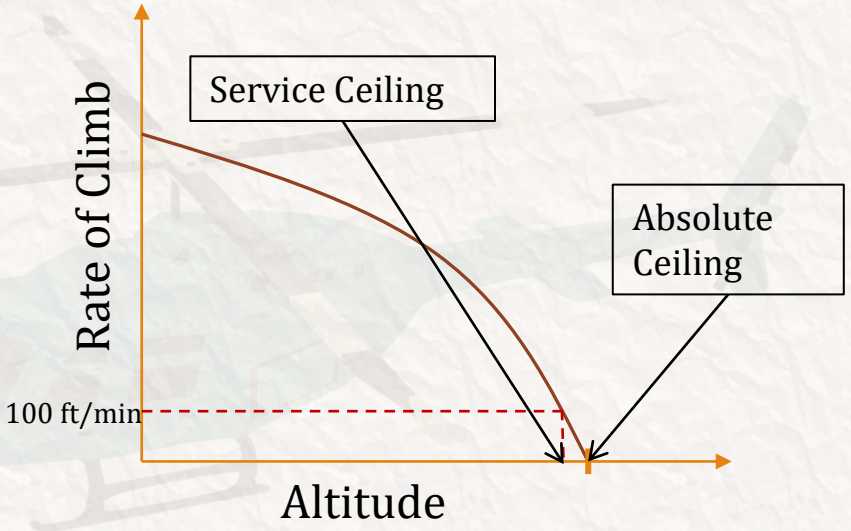
- Service ceiling ➔ Max climb rate = 100ft/min
- Absolute ceiling ➔ Max climb rate = 0

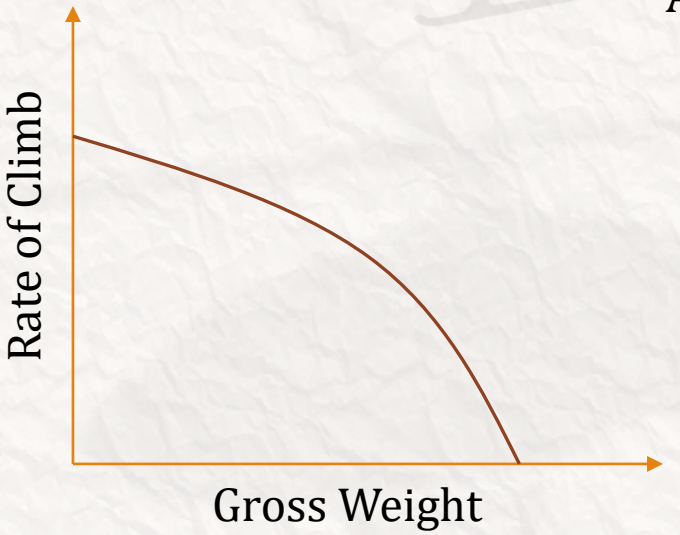
Power Balance in Hover/Climb (Graphical)



Climb performance trends







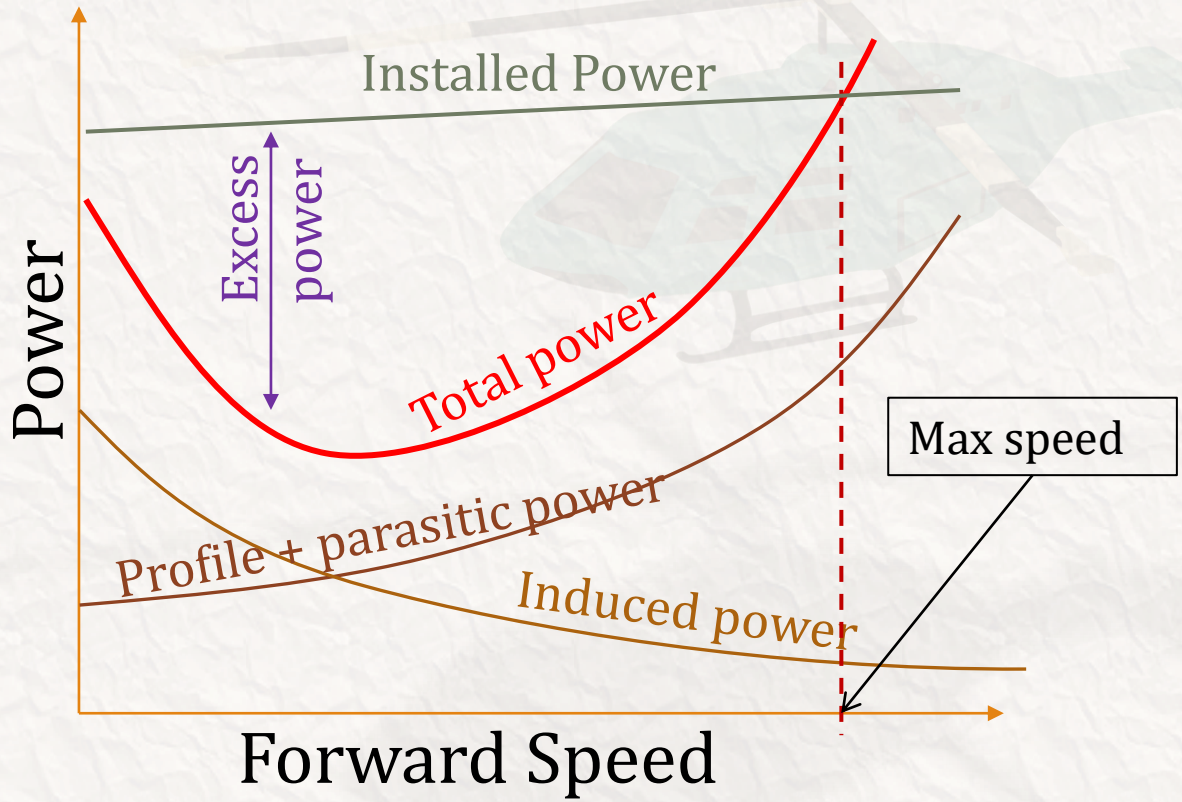
Forward flight performance

- Power requirement includes:
  - Main rotor power
    - Induced power (accounting for thrust = vehicle weight + downwash on fuselage) [P_i]
    - Profile power [P_o]
  - Tail rotor power (induced + profile power)
  - Fuselage parasite drag power (Drag × forward speed)
  - Engine power transmission and installation Losses

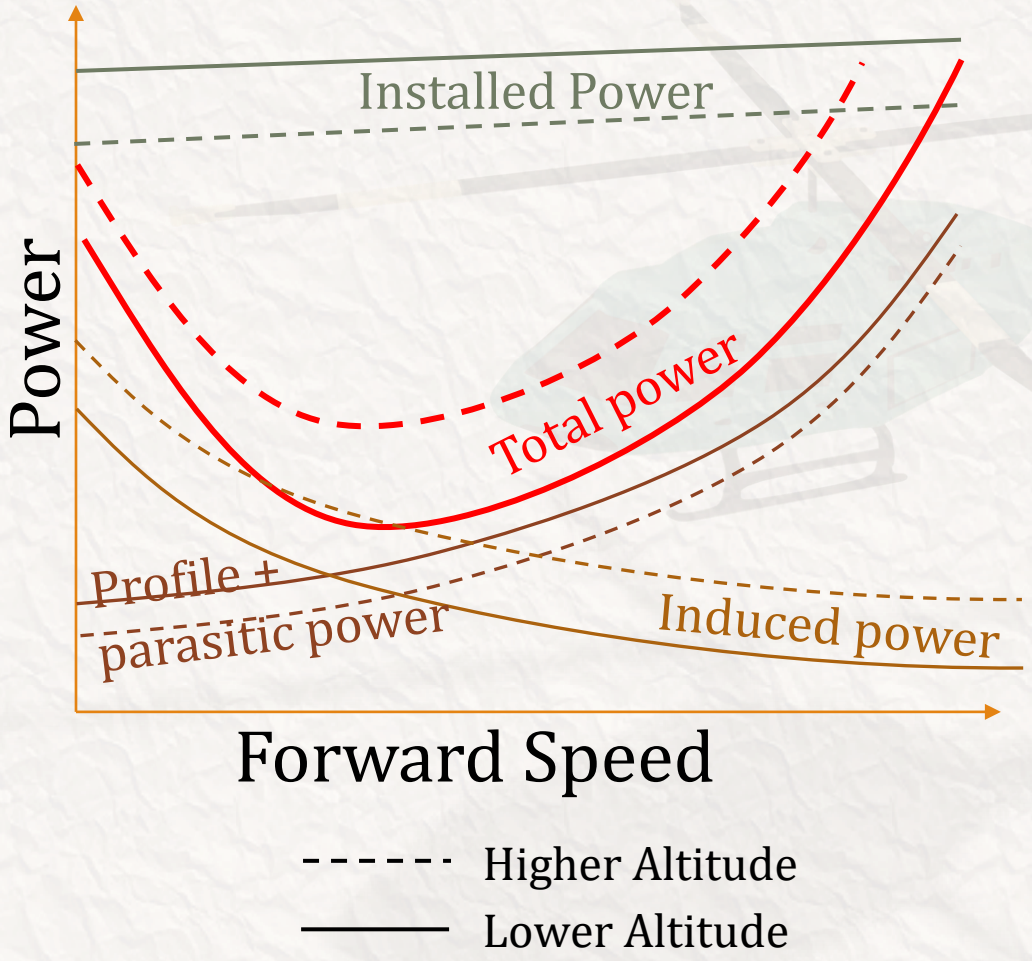
- Max speed is when:
  - Power required for forward flight = power available

- Climb rate = Excess power/Gross weight
  where excess power = installed power – power for forward flight

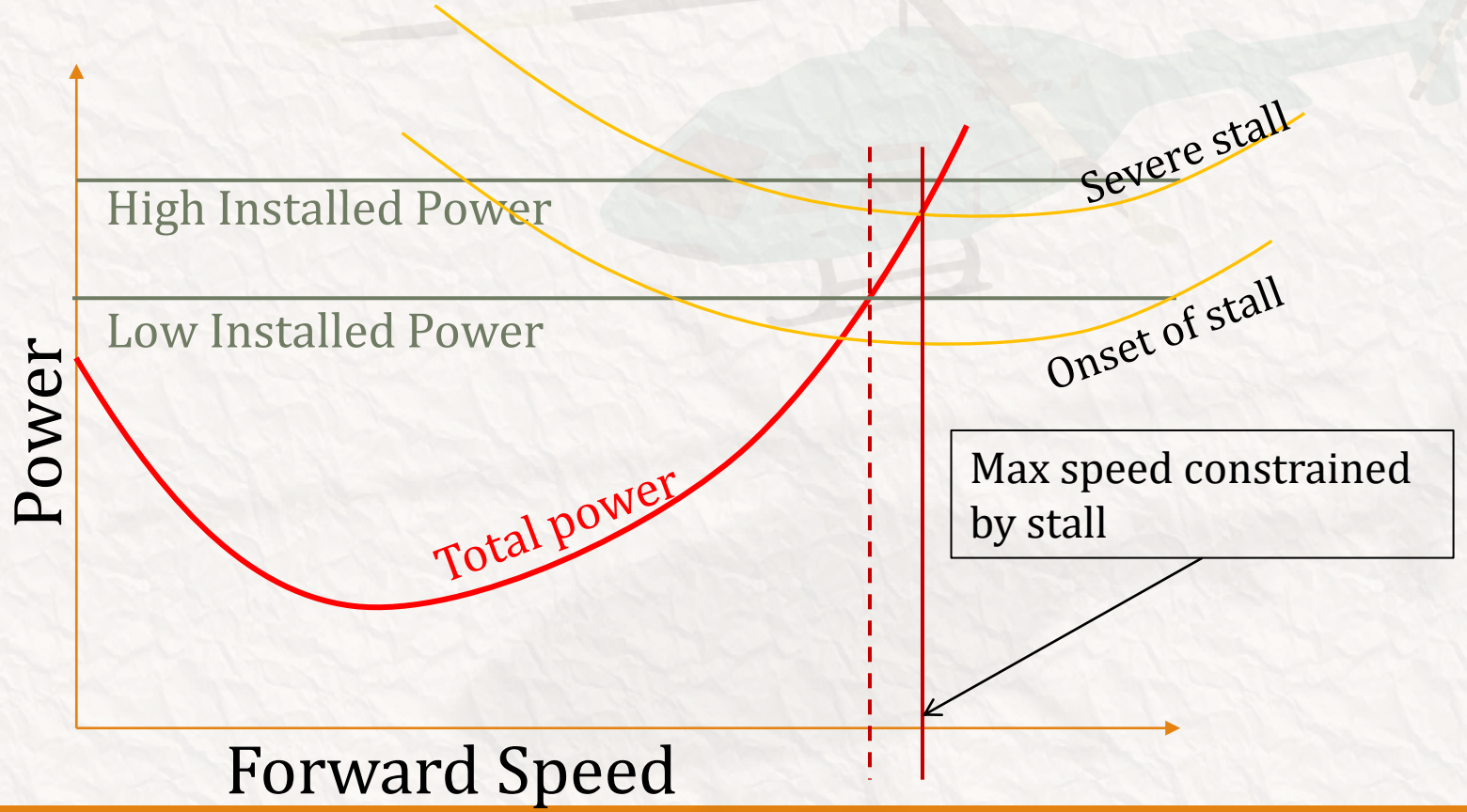
Power balance in level forward flight (Graphical)



Power balance in level forward flight at two different altitudes (Graphical)



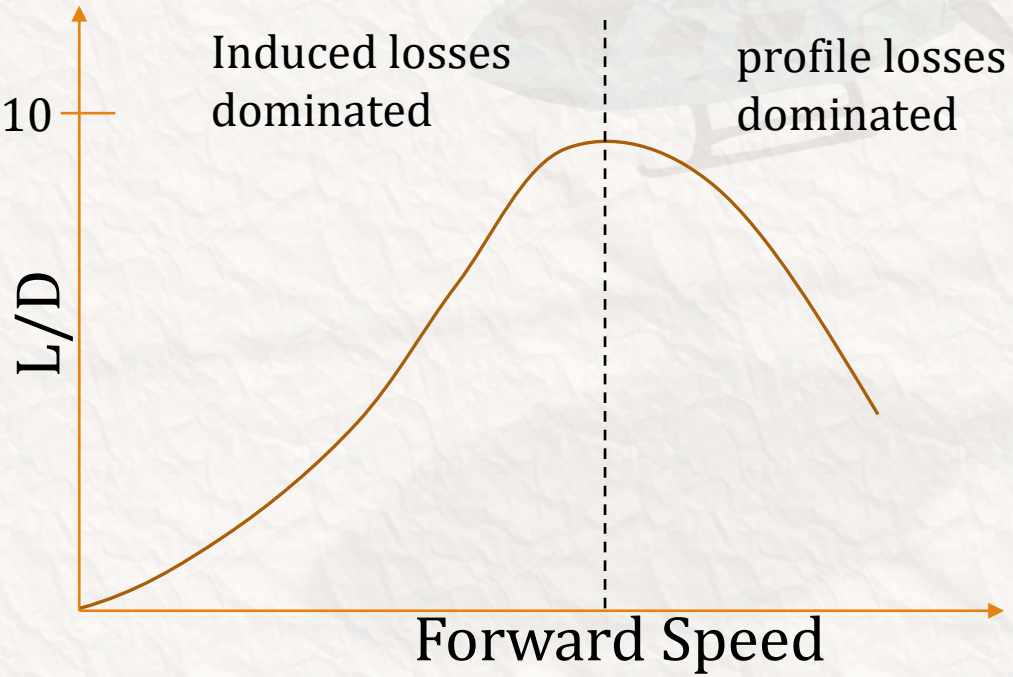
Max speed limited by stall



Equivalent Lift to Drag ratio for rotors

- L = Vertical component of rotor thrust
- $D = \frac{\text{Main rotor power}}{\text{forward speed}} - \text{Fuselage drag}$

  (only rotors are being compared, so fuselage drag subtracted)



Specific Range

- $Specific\ Range(SR) = \frac{Distance\ travelled}{Fuel\ burn}$
- $Fuel\ burn\ rate = Power \times fuel\ per\ kW$

Instantaneous specific range depends on fuel flow rate which is dependent on instantaneous power which is a function of instantaneous gross weight, altitude, speed, rotor geometry.

$$R = \int_{start}^{end} V dt = \int_{start}^{end} V \frac{-dW}{\text{fuel burn rate (kg/sec)}} = \int_{GW,landing}^{GW,take-off} (SR) d(GW)$$

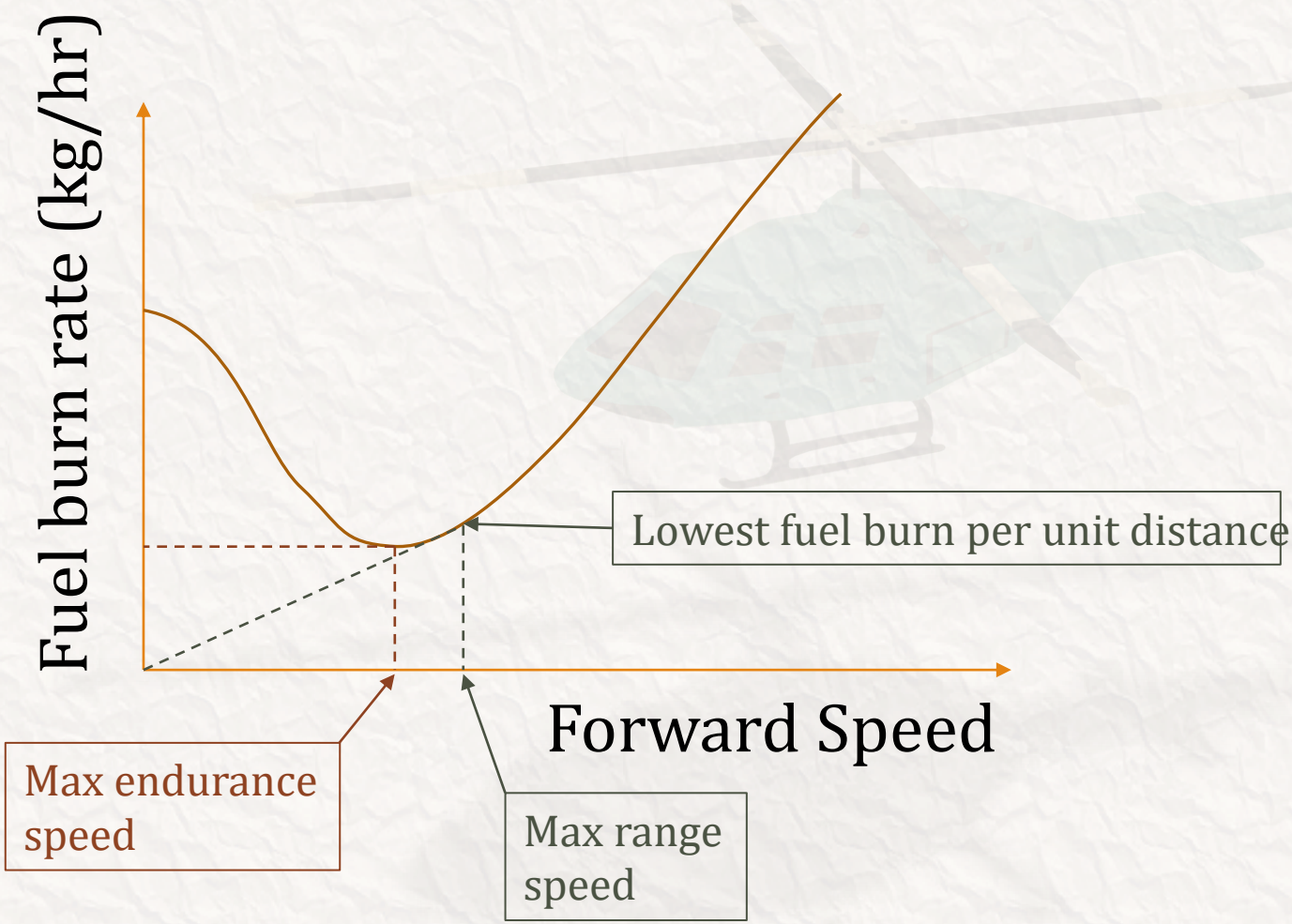
Endurance

- $Specific\ Endurance(SE) = \frac{1}{Fuel\ burn\ rate\ (kg/hr)}$
- $Fuel\ burn\ rate = Power\ @\ loiter \times fuel\ per\ kW$

$$\text{Total Endurance, hr} = \int_{\text{Start}}^{\text{End}} \text{dt}$$

$$= \int_{GWatlanding}^{GW,takeoff} (SE)d(GW)$$

Maximizing Range and Endurance



Payload Range

- Increase in payload decreases amount of fuel that can be carried for the given GW, reducing range

